

THE 3-CUT COMPLEX OF THE CIRCULAR LADDER IS A WEDGE OF SPHERES IN DEGREE $2n - 4$, FOR $n \geq 5$

ABSTRACT. Let $\Delta_k(G)$ be the k -cut complex of a graph G : the simplicial complex on $V(G)$ whose facets are the sets σ with $|\sigma| = |V(G)| - k$ for which $G[V(G) \setminus \sigma]$ is disconnected. Chauhan, Shukla and Vinayak conjectured that for $m = 2$ and $n \geq 5$ the reduced integral homology of $\Delta_3(K_m \square C_n)$ is $\mathbb{Z}^{2n^2-8n+1}$ in degree $mn - 4$ and zero elsewhere. We prove this, and add that for every $n \geq 5$ the complex $\Delta_3(C_n \square K_2)$ is homotopy equivalent to a wedge of $2n^2 - 8n + 1$ copies of S^{2n-4} . The proof is an integer linear algebra computation on the Alexander dual, which a parity argument confines to dimension 3 for every n , together with three explicit unimodular minors. The integer $2n^2 - 8n + 1$ and the vanishing of $\tilde{H}_i$ for $i \leq 2n - 7$ are already in print for this exact graph, as a reduced Euler characteristic and a codimension-2 skeleton, in work of Bayer, Denker, Jelić Milutinović, Rowlands, Sundaram and Xue. What we do not find in the literature cited here is the vanishing of the two groups those results leave free, the absence of torsion, and the homotopy type.

1. THE CONJECTURE

For a graph G on the vertex set V with $|V| = N$ and an integer $k \geq 1$, the k -cut complex $\Delta_k(G)$ is the simplicial complex on V whose facets are exactly the sets $\sigma \subseteq V$ with $|\sigma| = N - k$ such that the induced subgraph $G[V \setminus \sigma]$ is *disconnected* [1]. Equivalently,

- (1) $\sigma \in \Delta_k(G) \iff V \setminus \sigma$ contains a k -set T with $G[T]$ disconnected.

This is the non-total complex; the *total* k -cut complex, which demands that $G[V \setminus \sigma]$ have no edge at all, is a different object and is not considered here.

Chauhan, Shukla and Vinayak [4] state the following as the last numbered conjecture of their paper. The e-print carries no printed statement numbers, so we quote the statement in full rather than by number.

Conjecture 1 (Chauhan–Shukla–Vinayak). (i) For $m \geq 2$ and $n = 4$,

$$\tilde{H}_i(\Delta_3(K_m \square C_n)) = \begin{cases} \mathbb{Z}^{2m^2-5m+3} & \text{if } i = mn - 5, \\ \mathbb{Z}^{2m(m-1)} & \text{if } i = mn - 4, \\ 0 & \text{otherwise.} \end{cases}$$

$$(ii) \text{ For } m = 2 \text{ and } n \geq 5, \tilde{H}_i(\Delta_3(K_m \square C_n)) = \begin{cases} \mathbb{Z}^{2n^2-8n+1} & \text{if } i = mn - 4, \\ 0 & \text{otherwise.} \end{cases}$$

2020 *Mathematics Subject Classification.* 05E45, 55U10, 05C76.

Key words and phrases. cut complex, circular ladder, prism graph, Alexander duality, simplicial homology, wedge of spheres.

Part (ii) is the statement settled here. With $m = 2$ it reads: for every $n \geq 5$,

$$(2) \quad \tilde{H}_{2n-4}(\Delta_3(C_n \square K_2); \mathbb{Z}) \cong \mathbb{Z}^{2n^2-8n+1}, \quad \tilde{H}_i(\Delta_3(C_n \square K_2); \mathbb{Z}) = 0 \quad (i \neq 2n-4).$$

Homology is reduced and integral throughout, as in the source: Δ_3 is connected, so unreduced $H_0 \neq 0$ would contradict the clause “0 otherwise”.

2. THE RESULT

The Cartesian product is commutative up to isomorphism, so $K_2 \square C_n \cong C_n \square K_2$ and part (ii) of Conjecture 1 is a statement about the latter. Write $\text{CL}_n := C_n \square K_2$, the *circular ladder* or n -gonal prism: the vertex set is $V = \{(r, j) : r \in \{0, 1\}, j \in \mathbb{Z}_n\}$, with edges the *rails* $\{(r, j), (r, j+1)\}$ and the *rungs* $\{(0, j), (1, j)\}$. So $|V| = 2n$, $|E| = 3n$, and CL_n is 3-regular and triangle-free of girth 4. Write $Q_j := \{(0, j), (0, j+1), (1, j), (1, j+1)\}$ for the *rung square* at column j .

Theorem 2. *Let $n \geq 5$. Then*

$$\tilde{H}_i(\Delta_3(\text{CL}_n); \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{2n^2-8n+1} & i = 2n-4, \\ 0 & \text{otherwise,} \end{cases}$$

so Conjecture 1(ii) holds. Moreover $\Delta_3(\text{CL}_n)$ is homotopy equivalent to a wedge of $2n^2 - 8n + 1$ copies of S^{2n-4} .

Part of the content of Theorem 2 is published, and is due to Bayer, Denker, Jelić Milutinović, Rowlands, Sundaram and Xue. They prove [2] that if G has N vertices, $k \geq 3$, G has no cycle of length $\leq k$ and G has a cycle of length $k+1$, then $\Delta_k(G)$ contains a complete codimension-2 skeleton and

$$(3) \quad \tilde{\chi}(\Delta_k(G)) = (-1)^{N-k-1} \left(\binom{N-1}{k-1} + |Y_{k+1}(G)| - |Z_k(G)| \right),$$

where $Y_{k+1}(G)$ is the set of complements of $(k+1)$ -cycles and $Z_k(G)$ the set of complements of connected k -sets. The hypotheses hold verbatim for $G = \text{CL}_n$ and $k = 3$: girth $4 > 3$, and CL_n has 4-cycles. Substituting $N = 2n$, $|Y_4| = n$ and $|Z_3| = 6n$ (Lemma 3) into (3) gives $\tilde{\chi} = \binom{2n-1}{2} + n - 6n = 2n^2 - 8n + 1$. So the integer $2n^2 - 8n + 1$ is not new, and neither is the vanishing $\tilde{H}_i(\Delta_3(\text{CL}_n)) = 0$ for $i \leq 2n-7$, which is immediate from the complete codimension-2 skeleton.

Since $\Delta_3(\text{CL}_n)$ is pure of dimension $2n-4$, those two published facts leave three groups possibly nonzero — degrees $2n-6$, $2n-5$, $2n-4$ — constrained by one linear relation. What Theorem 2 adds is therefore the vanishing of $\tilde{H}_{2n-5}$ and $\tilde{H}_{2n-6}$, torsion-freeness, which (3) cannot see, and the homotopy type. We have not found those three statements in the literature cited here, and we make no priority claim: no comprehensive literature search was carried out.

3. THE PROOF

3.1. The Alexander dual is three-dimensional, for every n . For a simplicial complex K on V , $|V| = N$, the Alexander dual is $K^* = \{\sigma \subseteq V : V \setminus \sigma \notin K\}$, and combinatorial Alexander duality [3] gives

$$(4) \quad \tilde{H}_i(K^*; \mathbb{Z}) \cong \tilde{H}^{N-i-3}(K; \mathbb{Z}) \quad (i \in \mathbb{Z}).$$

By (1), $V \setminus \sigma \notin \Delta_3(G)$ if and only if every 3-subset of σ induces a connected subgraph. Hence

$$(5) \quad D := \Delta_3(G)^* = \{S \subseteq V : \text{every 3-subset of } S \text{ induces a connected subgraph}\},$$

which is manifestly a simplicial complex: a 3-subset of a subset of S is a 3-subset of S .

Lemma 3. *Let G be a cubic triangle-free graph on $N = 2n$ vertices with exactly q induced 4-cycles. Then D has f -vector*

$$(f_{-1}, f_0, f_1, f_2, f_3, f_4, \dots) = (1, 2n, \binom{2n}{2}, 6n, q, 0, 0, \dots),$$

so $\dim D = 3$ whenever $q > 0$. Moreover $q = n$ for $G = \text{CL}_n$ with $n \geq 5$, while $q = n + 2 = 6$ for $G = \text{CL}_4$.

Proof. Since G is triangle-free, a 3-set induces a connected subgraph precisely when it spans exactly two edges, that is, when it is an induced path P_3 : zero or one edge leaves an isolated vertex, and three edges would be a triangle.

A 2-set has no 3-subset, so $f_1 = \binom{2n}{2}$. A 3-set lies in D iff it is an induced P_3 , and the number of those is $\sum_v \binom{\deg v}{2} = 2n \binom{3}{2} = 6n$, with no triangle correction.

Let $|S| = s \geq 4$ with $S \in D$ and put $e = |E(G[S])|$. Each of the $\binom{s}{3}$ triples of S spans exactly 2 edges, and each edge of $G[S]$ lies in exactly $\binom{s-2}{1} = s-2$ of those triples. Counting edge-triple incidences twice,

$$(6) \quad (s-2)e = 2 \binom{s}{3}.$$

For $s = 5$, (6) reads $3e = 20$, which has no integer solution: $f_4 = 0$, and hence $f_j = 0$ for all $j \geq 4$ because D is a complex. For $s = 4$, (6) gives $e = 4$; a triangle-free graph on four vertices with four edges is C_4 , and $G[S]$ is an induced subgraph, so S spans an induced 4-cycle. Conversely every triple of an induced 4-cycle is an induced P_3 . So $f_3 = q$.

Finally, the 4-cycles of CL_n . A cycle of CL_n uses an even number of rungs. With no rung it lies in one rail, i.e. in a copy of C_n , which has a 4-cycle only if $n = 4$; the two rails are then induced 4-cycles, and this is the source of the two extra squares at $n = 4$. With two rungs, at columns j and j' , the cycle traverses a path of length $a \geq 1$ in one rail and a path of length $b \geq 1$ in the other, and $2 + a + b = 4$ forces $a = b = 1$, so $j' = j \pm 1$ and the cycle is a rung square Q_j ; all n of these are induced. With four or more rungs the

cycle has at least eight vertices. Hence $q = n$ for $n \geq 5$ and $q = n + 2$ for $n = 4$. $\square$

So for every $n \geq 5$ the simplicial chain complex of D is

$$(7) \quad 0 \longrightarrow \mathbb{Z}^n \xrightarrow{\partial_3} \mathbb{Z}^{6n} \xrightarrow{\partial_2} \mathbb{Z}^{n(2n-1)} \xrightarrow{\partial_1} \mathbb{Z}^{2n} \xrightarrow{\partial_0} \mathbb{Z} \longrightarrow 0,$$

a complex of *fixed length*, whose entries are the only thing that grows with n ; this is why a statement about all n is here a finite piece of linear algebra rather than an induction.

3.2. Three unimodular minors. We use the following standard fact: if an integer matrix M has rank r and some $r \times r$ minor of M equals ± 1 , then every invariant factor of M equals 1, because $d_1 d_2 \cdots d_r$ is the gcd of the $r \times r$ minors and $d_1 \mid d_2 \mid \cdots$. We exhibit such a minor for each of $\partial_1, \partial_2, \partial_3$.

Fix $n \geq 5$. Name the $6n$ triples of D by their centre: for $v = (r, j)$ put

$$\begin{aligned} T_1(v) &= \{(r, j-1), v, (r, j+1)\}, & T_2(v) &= \{(r, j-1), v, (1-r, j)\}, \\ T_3(v) &= \{(r, j+1), v, (1-r, j)\}. \end{aligned}$$

Since G is triangle-free, the centre of an induced P_3 is determined by its vertex set, so these $3 \cdot 2n = 6n$ triples are distinct and exhaust f_2 .

Lemma 4. $\text{rank } \partial_3 = n$ and every invariant factor of ∂_3 is 1.

Proof. The n generators of $C_3(D)$ are the squares Q_j . For $n \geq 5$ any two distinct squares meet in at most two vertices ($Q_j \cap Q_{j+1}$ is a rung, and $Q_j \cap Q_{j'} = \emptyset$ otherwise), so no triple lies in two squares and the n columns of ∂_3 have pairwise disjoint supports with entries ± 1 . Choosing one triple from each square gives an $n \times n$ minor equal to ± 1 ; the rank is therefore n . $\square$

Lemma 5. $\text{rank } \partial_1 = 2n - 1$ and every invariant factor of ∂_1 is 1.

Proof. $C_1(D)$ has all $\binom{2n}{2}$ pairs as generators, so ∂_1 is the incidence matrix of the complete graph K_{2n} . Its image lies in the sum-zero sublattice of $\mathbb{Z}^{2n}$, so $\text{rank } \partial_1 \leq 2n - 1$; the $2n - 1$ columns indexed by a spanning star give a $(2n - 1) \times (2n - 1)$ minor ± 1 after deleting the row of the star's centre. $\square$

Lemma 6. $\text{rank } \partial_2 = 5n$ and every invariant factor of ∂_2 is 1.

Proof. Upper bound. $\text{im } \partial_3 \subseteq \ker \partial_2$ and $\text{rank } \partial_3 = n$ by Lemma 4, so $\dim \ker \partial_2 \geq n$ and $\text{rank } \partial_2 \leq 6n - n = 5n$.

A $5n \times 5n$ minor equal to ± 1 . Select the $5n$ columns

$$\mathcal{C} = \{T_1(v) : v \in V\} \cup \{T_3(v) : v \in V\} \cup \{T_2((1, j)) : j \in \mathbb{Z}_n\}, \quad |\mathcal{C}| = 5n,$$

and the $5n$ rows

$$\begin{aligned} p_A(r, j) &= \{(r, j-1), (r, j+1)\}, & p_B(r, j) &= \{(r, j+1), (1-r, j)\}, \\ p_C(j) &= \{(1, j-1), (1, j)\}, \end{aligned}$$

for $r \in \{0, 1\}$, $j \in \mathbb{Z}_n$: respectively $2n$ same-rail pairs at distance 2, $2n$ diagonal pairs, and the n rail edges of row 1. For $n \geq 5$ these $5n$ pairs are distinct and of three distinct types, so they are $5n$ distinct rows.

We repeatedly use one observation. If two vertices of a triple $T \in D$ are non-adjacent then, T being an induced P_3 , they are its two endpoints and its third vertex is a common neighbour of them. So a non-adjacent pair p lies in exactly as many triples of D as the two vertices of p have common neighbours. For $n \geq 5$: the pair $p_A(r, j)$ has the single common neighbour (r, j) , so it lies only in $T_1((r, j))$; the pair $p_B(r, j)$ has the two common neighbours (r, j) and $(1 - r, j + 1)$, so it lies only in $T_3((r, j))$ and $T_2((1 - r, j + 1))$. (At $n = 4$ the first statement fails, since $(r, j - 1)$ and $(r, j + 1)$ then have two common neighbours; this is the second place the argument uses $n \geq 5$.) For the adjacent pair $p_C(j)$, the triples of D containing it are those obtained by adjoining a neighbour of one endpoint that is not adjacent to the other, namely $T_1((1, j - 1))$, $T_1((1, j))$, $T_3((1, j - 1))$ and $T_2((1, j))$.

Now delete rows and columns from the $5n \times 5n$ submatrix in four stages, at each stage using a row whose only surviving nonzero entry is ± 1 , so that the determinant is unchanged up to sign:

- (1) the $2n$ rows $p_A(r, j)$: each meets $\mathcal{C}$ only in $T_1((r, j))$. Delete them with the $2n$ columns $T_1(v)$.
- (2) the n rows $p_B(1, j)$: each lies in exactly two triples of D , namely $T_3((1, j))$ and $T_2((0, j + 1))$, and the latter is not in $\mathcal{C}$. Delete them with the n columns $T_3((1, j))$.
- (3) the n rows $p_C(j)$: of the four triples containing $p_C(j)$, three were deleted in stages 1 and 2, leaving $T_2((1, j))$. Delete them with the n columns $T_2((1, j))$.
- (4) the n rows $p_B(0, j)$: each meets $\mathcal{C}$ in $T_3((0, j))$ and $T_2((1, j + 1))$, and the latter was deleted in stage 3. Delete them with the n columns $T_3((0, j))$.

Nothing is left, so the minor is ± 1 and $\text{rank } \partial_2 \geq 5n$. $\square$

3.3. The homology of D , and of Δ_3 .

Proposition 7. *For $n \geq 5$, $\tilde{H}_i(D; \mathbb{Z}) = 0$ for $i \neq 1$, and $\tilde{H}_1(D; \mathbb{Z}) \cong \mathbb{Z}^{2n^2 - 8n + 1}$.*

Proof. By Lemmas 4–6 every invariant factor of ∂_1 , ∂_2 and ∂_3 equals 1, and ∂_0 is the augmentation, which is surjective with its single invariant factor 1. Since $\text{im } \partial_{i+1} \subseteq \ker \partial_i$ and $\ker \partial_i$ is free, the torsion subgroup of $\tilde{H}_i(D; \mathbb{Z})$ is the direct sum of the $\mathbb{Z}/d_j$ over the invariant factors d_j of ∂_{i+1} , hence trivial.

So every $\tilde{H}_i(D; \mathbb{Z})$ is free and it suffices to compute ranks:

$$\begin{aligned} \tilde{H}_0 : \quad (2n-1) - \text{rank } \partial_1 &= (2n-1) - (2n-1) = 0, \\ \tilde{H}_1 : \quad \binom{2n}{2} - \text{rank } \partial_1 - \text{rank } \partial_2 &= n(2n-1) - (2n-1) - 5n = 2n^2 - 8n + 1, \\ \tilde{H}_2 : \quad 6n - \text{rank } \partial_2 - \text{rank } \partial_3 &= 6n - 5n - n = 0, \\ \tilde{H}_3 : \quad n - \text{rank } \partial_3 &= n - n = 0. \end{aligned}$$

(For $\tilde{H}_0$ note $\ker \partial_0$ has rank $2n-1$.) All other $\tilde{H}_i$ vanish because $\dim D = 3$. $\square$

Proof of the homology statement of Theorem 2. By (4) with $N = 2n$ and $K = \Delta_3(\text{CL}_n)$, $K^* = D$, Proposition 7 gives $\tilde{H}^{2n-i-3}(\Delta_3; \mathbb{Z}) \cong \tilde{H}_i(D; \mathbb{Z})$, so $\tilde{H}^j(\Delta_3; \mathbb{Z}) = 0$ for every $j \neq 2n-4$ and $\tilde{H}^{2n-4}(\Delta_3; \mathbb{Z}) \cong \mathbb{Z}^{2n^2-8n+1}$.

Now transport to homology. Δ_3 is a finite complex, so all $\tilde{H}_i$ are finitely generated, and the universal coefficient theorem gives $\tilde{H}^j \cong \text{Hom}(\tilde{H}_j, \mathbb{Z}) \oplus \text{Ext}(\tilde{H}_{j-1}, \mathbb{Z})$, that is, the free part of $\tilde{H}_j$ direct sum the torsion of $\tilde{H}_{j-1}$. From $\tilde{H}^j = 0$ for $j \neq 2n-4$ we get: the free part of $\tilde{H}_j$ is 0 for every $j \neq 2n-4$, and the torsion of $\tilde{H}_{j-1}$ is 0 for every $j \neq 2n-4$, i.e. $\tilde{H}_i$ is torsion-free for every $i \neq 2n-5$. At $j = 2n-4$ the group $\tilde{H}^{2n-4} \cong \mathbb{Z}^{2n^2-8n+1}$ is free, and it contains the torsion of $\tilde{H}_{2n-5}$ as a direct summand, so that torsion is trivial too. Combining, every $\tilde{H}_i$ with $i \neq 2n-4$ is torsion-free with zero free part, hence zero, and $\tilde{H}_{2n-4} \cong \mathbb{Z}^{2n^2-8n+1}$. $\square$

3.4. The homotopy type.

Lemma 8. *For $n \geq 5$ every subset of V of size at most 3 is a face of $\Delta_3(\text{CL}_n)$; that is, $\Delta_3(\text{CL}_n)$ contains the full 2-skeleton of the simplex on V .*

Proof. Let $|\sigma| \leq 3$, so $|V \setminus \sigma| \geq 2n-3 \geq 7$. Pick $u \in V \setminus \sigma$. Among the $\geq 2n-4 \geq 6$ other vertices of $V \setminus \sigma$, at most 3 are neighbours of u , so at least $2n-7 \geq 3$ are not. Take two of them, w_1, w_2 : the triple $\{u, w_1, w_2\}$ spans at most one edge and so induces a disconnected subgraph. By (1), $\sigma \in \Delta_3(\text{CL}_n)$. $\square$

Proof of the homotopy statement of Theorem 2. Let $X = \Delta_3(\text{CL}_n)$ with $n \geq 5$, let $d = 2n-4 \geq 6$ and $r = 2n^2-8n+1$. By Lemma 8 the 2-skeleton of X is the 2-skeleton of a simplex, which is simply connected; since π_1 depends only on the 2-skeleton, X is simply connected. By the homology statement $\tilde{H}_*(X; \mathbb{Z})$ is free and concentrated in the single degree d .

We first record that X is $(d-1)$ -connected, which is what the Hurewicz theorem needs. Suppose X is $(j-1)$ -connected for some j with $2 \leq j \leq d-1$; since $j \geq 2$, Hurewicz gives $\pi_j(X) \cong \tilde{H}_j(X; \mathbb{Z}) = 0$, and hence X is j -connected. As X is simply connected, that is, 1-connected, induction from $j = 2$ gives $(d-1)$ -connectedness. Hurewicz applied once more, now in degree d , makes $\pi_d(X) \rightarrow H_d(X) \cong \mathbb{Z}^r$ an isomorphism. Choose maps $S^d \rightarrow X$ representing a basis and let $f : \bigvee_{i=1}^r S^d \rightarrow X$ be the resulting

map. Then f induces an isomorphism on integral homology in every degree, and both spaces are simply connected CW complexes, so f is a homotopy equivalence by the homology form of Whitehead's theorem [5]. Note $r = 2n^2 - 8n + 1 \geq 11 > 0$ for $n \geq 5$, so the wedge is non-empty. $\square$

4. SCOPE, AND VERIFICATION

Part (i) of Conjecture 1, which concerns $n = 4$ and $m \geq 2$, remains open; nothing here bears on $m \geq 3$. The hypothesis $n \geq 5$ is used twice, in Lemma 3 and in Lemma 6, and part (ii) is false at $n = 4$: an exact integral computation gives $\tilde{H}_3(\Delta_3(\text{CL}_4); \mathbb{Z}) \cong \mathbb{Z}$ and $\tilde{H}_4(\Delta_3(\text{CL}_4); \mathbb{Z}) \cong \mathbb{Z}^4$, and zero elsewhere.

The accompanying program `verify.py` (Python 3.9+, standard library only, exact integer arithmetic, no randomness) re-derives from the definitions: the graph invariants of CL_n and the f -vector of D for $n = 4, \dots, 40$, including the absence of a 5-set and the value $f_3 = 6$ at $n = 4$ (Lemma 3); the exact integral homology of $\Delta_3(\text{CL}_n)$ for $n = 5, \dots, 24$, together with the boundary ranks $n, 5n, 2n - 1$ of Lemmas 4–6 and the vanishing of all invariant factors above 1; the four-stage deletion of Lemma 6, executed as a stripping certificate on the $5n \times 5n$ submatrix itself for $n = 5, \dots, 24$; and the same homology computed *directly* from the full complex Δ_3 , with no duality and no relative complex, at $n = 4, 5, 6, 7$. Its recorded output is `verify.output.txt` (45 checks, all passing). Among the things the program states it does not cover are $n \geq 25$, any $m \geq 3$, the homotopy-theoretic step, and any literature search; for those, and for all $n \geq 5$, the paper rests on the proof above.

REFERENCES

- [1] M. Bayer, M. Denker, M. Jelić Milutinović, R. Rowlands, S. Sundaram, and L. Xue, *Topology of cut complexes of graphs*, SIAM J. Discrete Math. **38** (2024), no. 2, 1630–1675; arXiv:2304.13675.
- [2] M. Bayer, M. Denker, M. Jelić Milutinović, R. Rowlands, S. Sundaram, and L. Xue, *Topology of cut complexes II*, SIAM J. Discrete Math. **39** (2025), no. 2, 1123–1157. doi:10.1137/24M1676077; arXiv:2407.08158. The result used here is the one on graphs with no cycle of length $\leq k$ and at least one cycle of length $k + 1$; it is stated in full in Section 2 above.
- [3] A. Björner and M. Tancer, *Combinatorial Alexander duality—a short and elementary proof*, Discrete Comput. Geom. **42** (2009), no. 4, 586–593. doi:10.1007/s00454-008-9102-x.
- [4] P. Chauhan, S. Shukla, and K. Vinayak, *Total 2-cut complexes of powers of cycle graphs and Cartesian products of certain graphs*, arXiv:2512.04486v1, 2025. The conjecture settled here is the last numbered conjecture of that paper.
- [5] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.